N+ - Network Fundamentals and building network

Network Devices

[Q-1] Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

```
         Internet
            |
         [Router]
            |
         [Switch]
         /  |  \
        /   |   \
  [Laptop] [Smart TV] [Access Point]
                  |
             [Smartphone]
```

Connections

- **Router** → Connects the home network to the Internet.
- **Switch** → Connects multiple devices within the local network (LAN).
- **Laptop** → Connected to the switch using an Ethernet cable.
- **Smart TV** → Connected to the switch using an Ethernet cable.
- **Access Point** → Provides wireless (Wi-Fi) connectivity.
- **Smartphone** → Connected wirelessly to the Access Point through Wi-Fi

[Q-2] Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WIFI for streaming on YouTube, LAN gaming, etc.).

Device	Purpose	Real-World Use Case
Router	Connects a network to the Internet and routes data between networks	Using Wi-Fi to browse websites or watch YouTube
Switch	Connects multiple devices within a Local Area Network (LAN)	Connecting computers for LAN gaming or file sharing
Hub	Broadcasts incoming data to all connected devices	Small test or training network in a laboratory
Access Point (AP)	Provides wireless network access to devices	Connecting smartphones, tablets, and laptops to Wi-Fi

[Q-3] Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Scenario: Ordering Food on Zomato

When I order food on **Zomato** using my smartphone:

1. My smartphone sends the request through Wi-Fi to the **Access Point**.
2. The Access Point forwards the data to the **Switch**.
3. The Switch sends the data to the **Router**.
4. The Router forwards the request to the Internet.
5. The request reaches Zomato's servers.
6. The server sends a response back through the same path to my smartphone.

Roles of Network Devices

- **Access Point (AP):** Provides wireless connectivity and allows smartphones to connect to the network using Wi-Fi.
- **Switch:** Connects devices within the local area network (LAN) and forwards data to the correct device.
- **Router:** Connects the home network to the Internet and routes data between the local network and external networks.

[Q-4] Given this situation: you have a WIFI router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WIFI extenders.

1. Powerline Adapter

How it works:

- A pair of powerline adapters uses your home's electrical wiring to transmit network data.
- Connect one adapter to the router using an Ethernet cable and plug it into a power outlet.
- Plug the second adapter into a power outlet near the gaming PC and connect it to the PC with an Ethernet cable.

2. Wi-Fi Extender (with Ethernet Port)

How it works:

- The Wi-Fi extender connects wirelessly to the router and extends the Wi-Fi signal to a distant area.
- Connect the gaming PC to the extender's Ethernet port using an Ethernet cable.
- The extender acts as a bridge between the PC and the Wi-Fi network.